

Data Management Technologies

Unit 4 – Consolidation Task 1

Introduction

NoSQL (Not Only SQL) databases can store data with structures beyond the row/column structure of traditional relational databases.

Advantages of NoSQL databases

1. **Availability of different formats:** NoSQL databases can store data that differs from the tabular or relational format. NoSQL databases might, for example, feature key-value pairs in a JSON file.
As such, NoSQL is more dynamic and can apply well in different scenarios.
2. **Horizontal scaling is easier in NoSQL databases:** This means that increased loads can be more easily distributed through the addition of more machines.
3. **Consistent hashing:** This makes it easier to add and remove machines from the server in a non-disruptive manner. Servers and keys are placed on a circular ring and are responsible for the key between themselves and the next server, working anticlockwise to the server.
When a server is added or removed, only the servers nearest to them are affected, so extensive redistribution is avoided.
4. **NoSQL databases follow the BASE data consistency model:** This entails:
 - **Basic Availability** - data retrieval works most of the time.
 - **Soft state** - write consistency is not guaranteed, even after the write (inputted data) is acknowledged, as data might be propagating across nodes.
 - **Eventually Consistent** - even though nodes are not always consistent, they can become consistent after a period with no new updates.

The basic availability and eventually consistent aspects of the model are key advantages.

Disadvantages of NoSQL databases

1. **NoSQL databases do not follow the ACID data consistency model:** ACID guarantees include **Atomicity** (no half-finished operations), **Consistency** (only valid data states

should exist), **Isolation** (transactions do not interfere with each other), and **Durability** (completed transactions persist, even if the server crashes)

2. **Sync time lags:** If synchronisation is not fast enough, two users might see different versions of the same data.
3. **No schema enforcement:** Rows need not match a column's data structure. As such, inconsistent data types can be entered.
4. **NoSQL databases are incompatible with data that must be in a relational (tabular) structure:** Relational data is likely better handled with an SQL database.
5. **NoSQL queries are more complex than SQL queries:** As such, learning to add and update data on NoSQL databases is more difficult.

Applications

SQL Database

- **Library book management system.** The same fields of information will be stored for each book. For example: title, format, author, publisher, edition, year published, genre, availability, condition and shelf location.
Each field will have a fixed data type, for example, string for title and integer for edition. The database can also feature enumerated data types for availability and genre.
- **Financial transactions:** An SQL database will work well for storing records of financial transactions, for example, for a bank or for the finance department of a business. It has the advantage of consistency (within ACID), meaning that data is promptly updated, and users view the same version of data.

NoSQL Database

- **Physical sensor data:** A NoSQL database will work well for collecting data from physical sensors, for example, devices that collect inputs in the form of temperature or speed. A NoSQL database can more easily handle inputs from a variety of types of devices, as they easily accommodate various data structures and inputs of different data types.

- **Gaming Leaderboards:** These include simple key-value pairs (player-score fields) and change rapidly, with users moving up or down the board. Leaderboards do not necessarily require complex joins with external data. With a NoSQL database, users can be easily added and removed. As scores change, they can be read and written to the database swiftly.

References

IBM. (2022). *What is a NoSQL database?* IBM Think. <https://www.ibm.com/think/topics/nosql-databases>

MongoDB. (2026). *Advantages of NoSQL databases*. Mongo DB. <https://www.mongodb.com/resources/basics/databases/nosql-explained/advantages>